Fiscal Year:	FY 2017	**Task Last Updated:**	FY 10/13/2017
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Free Radical Theory of Aging in Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	Ruth.K.Globus@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	08/01/2016	**End Date:**	07/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: Change in period of performance to 8/1/2016-7/31/2018 (originally 11/1/2014-10/31/2017) per F. Hernandez/ARC (Ed., 7/29/16)		
Key Personnel Changes/Previous PI:	October 2017 report: New Co-Investigators: Candice Tahimic (NASA Ames Research Center); David Goukassian (Temple University). Remove Co-Investigator: Mohammed Shahnazari.		
COI Name (Institution):	Alwood, Joshua Ph.D. (NASA Ames Research Center) Berkowitz, Daniel M.D., Ph.D. (Johns Hopkins School of Medicine) Limoli, Charles Ph.D. (University of California, Irvine) Santhanam, Lakshmi Ph.D. (Johns Hopkins University) Schreurs, Ann-Sofie Ph.D. (NASA Ames Research Center) Tahimic, Candice GT Ph.D. (NASA Ames Research Center) Goukassian, David M.D., Ph.D. (Temple University)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Many of the adverse physiological effects of spaceflight on astronauts, such as cardiovascular and musculoskeletal deconditioning, resemble those of aging on Earth, leading to the hypothesis that reduced gravity accelerates human aging. This idea is based on numerous spaceflight findings at the organ-system level observed in both humans and animals. Possible specific, mechanistic links between aging and gravity at the molecular level have yet to be established; this is the knowledge gap that we will address in the proposed research project. Our Specific Aims will test responses to long duration spaceflight of a range of tissues (i.e., neural, cardiac, skeletal) in both wild type (control) and transgenic mice. The transgenic mice are engineered to have less oxidative damage in mitochondria, which may occur as a consequence of exposure to long duration spaceflight as well as progressive aging. Female mice (both wildtype and transgenic) will be exposed to spaceflight or kept as ground controls for as long as two months, then tissues (neural, heart, bone) recovered for subsequent analysis of gene expression, oxidative stress biomarkers, and tissue functions. Thus, results from this research will identify basic molecular mechanisms that may underlie degenerative changes arising during prolonged spaceflight or progressive aging on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this research will identify basic molecular mechanisms that may underlie degenerative changes arising during prolonged spaceflight or progressive aging on Earth.
Task Progress:	The free radical theory of aging was first formulated by Harman in the 1950s (Harman, 1956) and attributes the consequences of aging to the accumulation of oxidative damage due to excess free radicals (Finkel and Holbrook, 2000). The spaceflight environment including microgravity, ionizing radiation, altered nutrition, confinement, and inactivity poses multiple challenges to homeostasis. These factors may contribute to cellular stress which in turn may lead to increased generation of reactive oxygen species (ROS), DNA damage, and aberrant repair, cell cycle arrest, cell senescence, or death. Alterations in gene expression and persistent DNA damage may also contribute to tissue dysfunction over the long term. Potential mechanistic links between aging and spaceflight at the molecular level have yet to be established. Therefore, the main goal of this project is to establish the importance of cellular redox defense for physiological adaptation and tissue degeneration in the space environment. To accomplish this, we are using use both wildtype (WT) mice and a well-established, genetically-engineered animal model (mCAT mice), which exhibits extended lifespan via quenching mitochondrial ROS (Schriner et al., 2005). The animal model selected to test these ideas is genetically engineered to reduce reactive oxygen species (ROS) in mitochondria by targeted expression of the human catalase gene to the mitochondrial matrix (mCAT). Thus, mCAT mice provide a novel opportunity to conduct the first proof-of-principle test of the free radical theory of aging in space. Our central hypothesis is that prolonged exposure to the spaceflight environment leads to excess production of ROS and oxidative damage, culminating in accelerated tissue degeneration of multiple tissues (CNS, skeletal, cardiovascular). Our previous studies revealed that mCAT mice display reduced cognitive dysfunction after irradiation compared to wildtype mice (Liao et al., 2013). Our preliminary results indicate that mCAT mice displayed significantly increased catalase enzymatic activity (3-4 fold) compared to WT mice in bone, osteoblast cultures, and in osteoclast precursors; thereby confirming the transgene is enzymatically active in bone. After a treatment known to cause oxidative damage and tissue degeneration (i.e., 14 days of hindlimb unloading to simulate weightlessness and total body irradiation with 137Cs, 2Gy), mCAT mice displayed reduced oxidative damage markers compared to WT mice as indicated by reductions in levels of lipid peroxidation. In terms of skeletal structure, mCAT mice displayed significant differences compared to WT mice in several cancellous and cortical parameters. Together, these results suggest a role of mitochondrial ROS in skeletal growth and remodeling. By conducting studies on Earth and in space, we expect to gain insight into possible future intervention strategies for improving spaceflight adaptation and recovery. References Harman D. Aging: a theory based on free radical and radiation chemistry. J Gerontol. 1956 Jul;11(3):298-300. Finkel T, Holbrook NJ. Oxidants, oxidative stress and the biology of ageing. Nature. 2000 Nov 9;408(6809):239-47. Schriner SE, Linford NJ, Martin GM, Treuting P, Ogburn CE, Emond M, Coskun PE, Ladiges W, Wolf N, Van Remmen H, Wallace DC, Rabinovitch PS. Extension of murine life span by overexpression of catalase targeted to mitochondria. Science. 2005 Jun 24;308(5730):1909-11. Liao AC, Craver BM, Tseng BP, Tran KK, Parihar VK, Acharya MM, Limoli CL. Mitochondrial-targeted human catalase affords neuroprotection from proton irradiation. Radiat Res. 2013 Jul;180(1):1-6.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Globus RKG, Tahmic CGT, Torres AS, Alwood AS, Almeida E, Goukassian D, Santhanam L, Berkowitz D, Limoli C, Schreurs AS. "Free Radical Theory of Aging in Space." Presented at the 2017 International Space Station (ISS) Research and Development Conference, Washington, DC, July 17-20, 2017. 2017 International Space Station (ISS) Research and Development Conference, Washington, DC, July 17-20, 2017. , Jul-2017
Articles in Peer-reviewed Journals	Tahimic CGT, Globus RK. "Redox signaling and its impact on skeletal and vascular responses to spaceflight." International Journal of Molecular Sciences. Special Issue: Oxidative Stress and Space Biology: An Organ-Based Approach. 2017;18(10):2153. Published: 16 October 2017. https://doi.org/10.3390/ijms18102153 , Oct-2017